

Simplification with 4 variables. Many small implicants

Exercise 1

ab / cd	00	01	11	10
00	1	1	0	1
01	0	1	0	1
11	1	0	0	1
10	1	1	1	0

Exercise 2

ab / cd	00	01	11	10
00	1	1	0	0
01	0	1	0	1
11	0	1	1	1
10	0	0	0	0

Exercise 3

ab / cd	00	01	11	10
00	1	1	0	1
01	0	0	0	1
11	1	0	0	0
10	0	1	1	0

Exercise 4

ab / cd	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	0	1	1	0
10	1	1	0	0

Exercise 5

ab / cd	00	01	11	10
00	1	0	0	0
01	1	0	1	1
11	0	1	0	1
10	1	0	1	1

Exercise 6

ab / cd	00	01	11	10
00	1	1	0	0
01	0	1	0	1
11	0	1	1	0
10	1	0	1	0

Exercise 7

ab / cd	00	01	11	10
00	1	1	1	0
01	0	0	1	1
11	0	0	0	1
10	0	1	1	1

Exercise 8

ab / cd	00	01	11	10
00	1	1	1	1
01	0	1	0	1
11	1	1	1	0
10	1	0	1	0

Exercise 9

ab / cd	00	01	11	10
00	0	1	1	0
01	1	0	0	0
11	1	0	1	0
10	1	0	1	1

Exercise 10

ab / cd	00	01	11	10
00	0	0	1	1
01	0	1	1	0
11	1	0	0	1
10	0	1	0	1

Exercise 11

ab / cd	00	01	11	10
00	1	0	1	1
01	1	1	0	0
11	0	1	1	1
10	1	1	1	0

Exercise 12

ab / cd	00	01	11	10
00	0	0	1	1
01	1	1	1	0
11	0	1	0	1
10	1	1	0	1

Exercise 13

ab / cd	00	01	11	10
00	1	0	0	0
01	0	0	1	0
11	1	0	0	1
10	1	1	1	0

Exercise 14

ab / cd	00	01	11	10
00	1	0	0	0
01	1	0	1	0
11	1	0	1	1
10	0	0	1	0

Exercise 1

ab / cd	00	01	11	10
00	1	1	0	1
01	0	1	0	1
11	1	0	0	1
10	1	1	1	0

Solution to Exercise 1

There are 8 prime implicants:

$$\begin{aligned}
 &\bar{b}\bar{c} \\
 &a\bar{b}d \\
 &\bar{a}\bar{c}d \\
 &ab\bar{d} \\
 &\bar{a}\bar{b}\bar{d} \\
 &\bar{a}c\bar{d} \\
 &bc\bar{d} \\
 &a\bar{c}\bar{d}
 \end{aligned}$$

A minimum cover is made of 5 prime implicants:

$$a\bar{b}d + \bar{a}\bar{c}d + \bar{a}\bar{b}\bar{d} + bc\bar{d} + a\bar{c}\bar{d}$$

Exercise 2

ab / cd	00	01	11	10
00	1	1	0	0
01	0	1	0	1
11	0	1	1	1
10	0	0	0	0

Solution to Exercise 2

There are 6 prime implicants:

$$\begin{aligned}
 &abc \\
 &\bar{a}\bar{b}\bar{c} \\
 &abd \\
 &\bar{a}\bar{c}d \\
 &b\bar{c}d \\
 &bc\bar{d}
 \end{aligned}$$

A minimum cover is made of 4 prime implicants:

$$\bar{a}\bar{b}\bar{c} + bc\bar{d} + abd + b\bar{c}d$$

Exercise 3

ab / cd	00	01	11	10
00	1	1	0	1
01	0	0	0	1
11	1	0	0	0
10	0	1	1	0

Solution to Exercise 3

There are 6 prime implicants:

$$\bar{a}\bar{b}\bar{c}$$

$$a\bar{b}d$$

$$\bar{b}\bar{c}d$$

$$\bar{a}\bar{b}\bar{d}$$

$$\bar{a}c\bar{d}$$

$$ab\bar{c}\bar{d}$$

A minimum cover is made of 4 prime implicants:

$$a\bar{b}d + \bar{a}c\bar{d} + ab\bar{c}\bar{d} + \bar{a}\bar{b}\bar{c}$$

Exercise 4

ab / cd	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	0	1	1	0
10	1	1	0	0

Solution to Exercise 4

There are 8 prime implicants:

$$\bar{a}\bar{b}c$$

$$a\bar{b}\bar{c}$$

$$abd$$

$$a\bar{c}d$$

$$b\bar{c}d$$

$$\bar{a}\bar{b}\bar{d}$$

$$\bar{a}c\bar{d}$$

$$\bar{b}\bar{c}\bar{d}$$

A minimum cover is made of 6 prime implicants:

$$\bar{a}\bar{b}c + ab\bar{d} + b\bar{c}d + \bar{a}c\bar{d} + a\bar{b}\bar{c} + \bar{b}\bar{c}\bar{d}$$

Exercise 5

ab / cd	00	01	11	10
00	1	0	0	0
01	1	0	1	1
11	0	1	0	1
10	1	0	1	1

Solution to Exercise 5

There are 9 prime implicants:

$$\bar{a}bc$$

$$a\bar{b}c$$

$$\bar{a}b\bar{d}$$

$$a\bar{b}\bar{d}$$

$$ac\bar{d}$$

$$bc\bar{d}$$

$$\bar{a}\bar{c}\bar{d}$$

$$\bar{b}\bar{c}\bar{d}$$

$$ab\bar{c}d$$

A minimum cover is made of 6 prime implicants:

$$\bar{a}bc + a\bar{b}c + ab\bar{c}d + bc\bar{d} + \bar{a}\bar{c}\bar{d} + \bar{b}\bar{c}\bar{d}$$

Exercise 6

ab / cd	00	01	11	10
00	1	1	0	0
01	0	1	0	1
11	0	1	1	0
10	1	0	1	0

Solution to Exercise 6

There are 7 prime implicants:

$$\begin{aligned} &\bar{a}\bar{b}\bar{c} \\ &abd \\ &acd \\ &\bar{a}\bar{c}d \\ &b\bar{c}d \\ &\bar{b}\bar{c}\bar{d} \\ &\bar{a}bc\bar{d} \end{aligned}$$

A minimum cover is made of 5 prime implicants:

$$acd + \bar{b}\bar{c}\bar{d} + \bar{a}bc\bar{d} + \bar{a}\bar{c}d + b\bar{c}d$$

Exercise 7

ab / cd	00	01	11	10
00	1	1	1	0
01	0	0	1	1
11	0	0	0	1
10	0	1	1	1

Solution to Exercise 7

There are 7 prime implicants:

$$\begin{aligned} &\bar{b}d \\ &\bar{a}bc \\ &a\bar{b}c \\ &\bar{a}\bar{b}\bar{c} \\ &\bar{a}cd \\ &ac\bar{d} \\ &bc\bar{d} \end{aligned}$$

A minimum cover is made of 4 prime implicants:

$$\bar{b}d + \bar{a}\bar{b}\bar{c} + \bar{a}bc + ac\bar{d}$$

Exercise 8

ab / cd	00	01	11	10
00	1	1	1	1
01	0	1	0	1
11	1	1	1	0
10	1	0	1	0

Solution to Exercise 8

There are 10 prime implicants:

$$\begin{aligned}
 &\bar{a}\bar{b} \\
 &a b \bar{c} \\
 &a b d \\
 &a c d \\
 &\bar{b} c d \\
 &\bar{a} \bar{c} d \\
 &b \bar{c} d \\
 &\bar{a} c \bar{d} \\
 &a \bar{c} \bar{d} \\
 &\bar{b} \bar{c} \bar{d}
 \end{aligned}$$

A minimum cover is made of 5 prime implicants:

$$\bar{a} c \bar{d} + \bar{a} \bar{b} + a c d + b \bar{c} d + a \bar{c} \bar{d}$$

Exercise 9

ab / cd	00	01	11	10
00	0	1	1	0
01	1	0	0	0
11	1	0	1	0
10	1	0	1	1

Solution to Exercise 9

There are 7 prime implicants:

$$\begin{aligned}
 &a \bar{b} c \\
 &\bar{a} \bar{b} d \\
 &a c d \\
 &\bar{b} c d \\
 &a \bar{b} \bar{d}
 \end{aligned}$$

$$a\bar{c}\bar{d}$$

$$b\bar{c}\bar{d}$$

A minimum cover is made of 4 prime implicants:

$$\bar{a}\bar{b}d + a\bar{c}d + b\bar{c}\bar{d} + a\bar{b}\bar{d}$$

Exercise 10

ab / cd	00	01	11	10
00	0	0	1	1
01	0	1	1	0
11	1	0	0	1
10	0	1	0	1

Solution to Exercise 10

There are 7 prime implicants:

$$\bar{a}\bar{b}c$$

$$\bar{a}bd$$

$$\bar{a}cd$$

$$ab\bar{d}$$

$$ac\bar{d}$$

$$\bar{b}c\bar{d}$$

$$a\bar{b}\bar{c}d$$

A minimum cover is made of 5 prime implicants:

$$\bar{a}bd + ab\bar{d} + a\bar{b}\bar{c}d + \bar{a}\bar{b}c + \bar{b}c\bar{d}$$

Exercise 11

ab / cd	00	01	11	10
00	1	0	1	1
01	1	1	0	0
11	0	1	1	1
10	1	1	1	0

Solution to Exercise 11

There are 10 prime implicants:

$$ad$$

$$abc$$

$$\bar{a}\bar{b}c$$

$$\bar{a}b\bar{c}$$

$$a\bar{b}\bar{c}$$

$$\bar{b}cd$$

$$b\bar{c}d$$

$$\bar{a}\bar{b}\bar{d}$$

$$\bar{a}\bar{c}\bar{d}$$

$$\bar{b}\bar{c}\bar{d}$$

A minimum cover is made of 5 prime implicants:

$$abc + ad + \bar{a}\bar{b}c + \bar{a}b\bar{c} + \bar{b}\bar{c}\bar{d}$$

Exercise 12

ab / cd	00	01	11	10
00	0	0	1	1
01	1	1	1	0
11	0	1	0	1
10	1	1	0	1

Solution to Exercise 12

There are 10 prime implicants:

$$\bar{a}\bar{b}c$$

$$\bar{a}b\bar{c}$$

$$a\bar{b}\bar{c}$$

$$\bar{a}bd$$

$$\bar{a}cd$$

$$a\bar{c}d$$

$$b\bar{c}d$$

$$a\bar{b}\bar{d}$$

$$a\bar{c}\bar{d}$$

$$\bar{b}\bar{c}\bar{d}$$

A minimum cover is made of 6 prime implicants:

$$\bar{a}b\bar{c} + a\bar{c}\bar{d} + \bar{a}\bar{b}c + a\bar{b}\bar{c} + \bar{a}cd + a\bar{c}d$$

Exercise 13

ab / cd	00	01	11	10
00	1	0	0	0
01	0	0	1	0
11	1	0	0	1
10	1	1	1	0

Solution to Exercise 13

There are 6 prime implicants:

$$a\bar{b}\bar{c}$$

$$a\bar{b}d$$

$$ab\bar{d}$$

$$a\bar{c}\bar{d}$$

$$\bar{b}\bar{c}\bar{d}$$

$$\bar{a}bcd$$

A minimum cover is made of 4 prime implicants:

$$a\bar{b}d + ab\bar{d} + \bar{b}\bar{c}\bar{d} + \bar{a}bcd$$

Exercise 14

ab / cd	00	01	11	10
00	1	0	0	0
01	1	0	1	0
11	1	0	1	1
10	0	0	1	0

Solution to Exercise 14

There are 6 prime implicants:

$$abc$$

$$acd$$

$$bcd$$

$$ab\bar{d}$$

$$\bar{a}\bar{c}\bar{d}$$

$$b\bar{c}\bar{d}$$

A minimum cover is made of 4 prime implicants:

$$acd + bcd + \bar{a}\bar{c}\bar{d} + ab\bar{d}$$